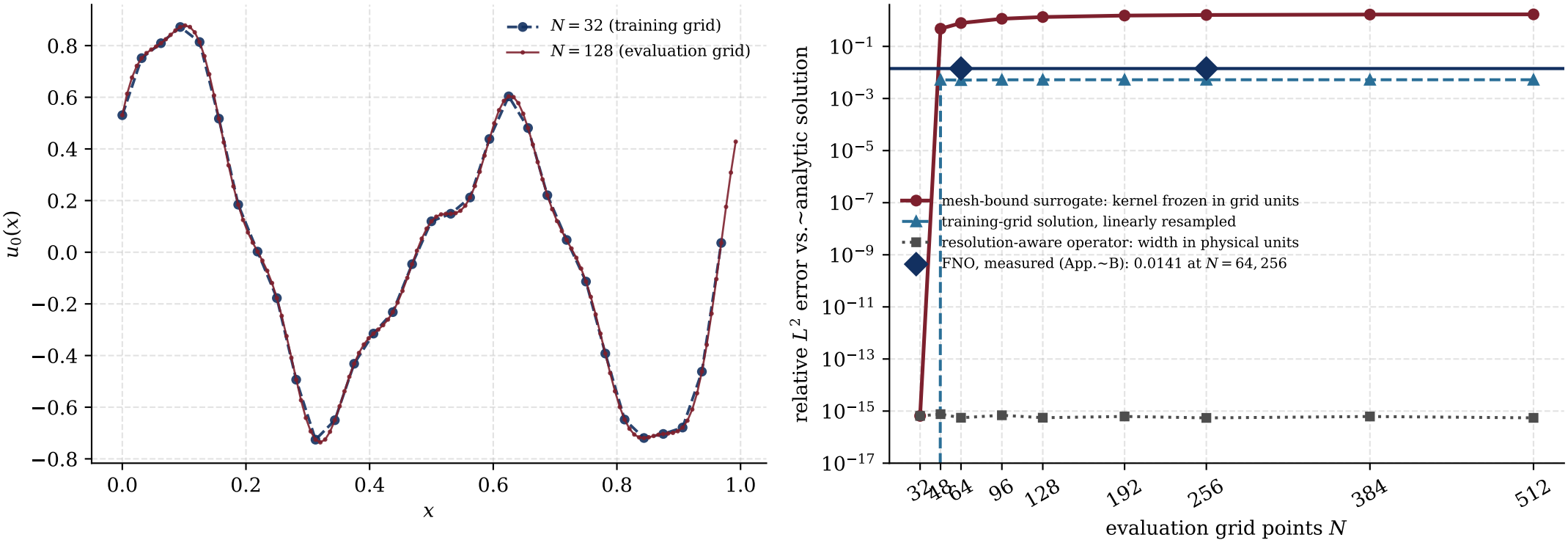
(a) One continuous field, two discretizations
$N = 32$ (training grid)
$N = 128$ (evaluation grid)
$u_0(x)$
x
(b) Error vs resolution, calibrated at $N = 32$
relative L^2 error vs.~analytic solution
mesh-bound surrogate: kernel frozen in grid units
training-grid solution, linearly resampled
resolution-aware operator: width in physical units
FNO, measured (App.~B): 0.0141 at $N = 64, 256$
evaluation grid points N